Week 1 Case Study

Analysis of Myntra Apparel

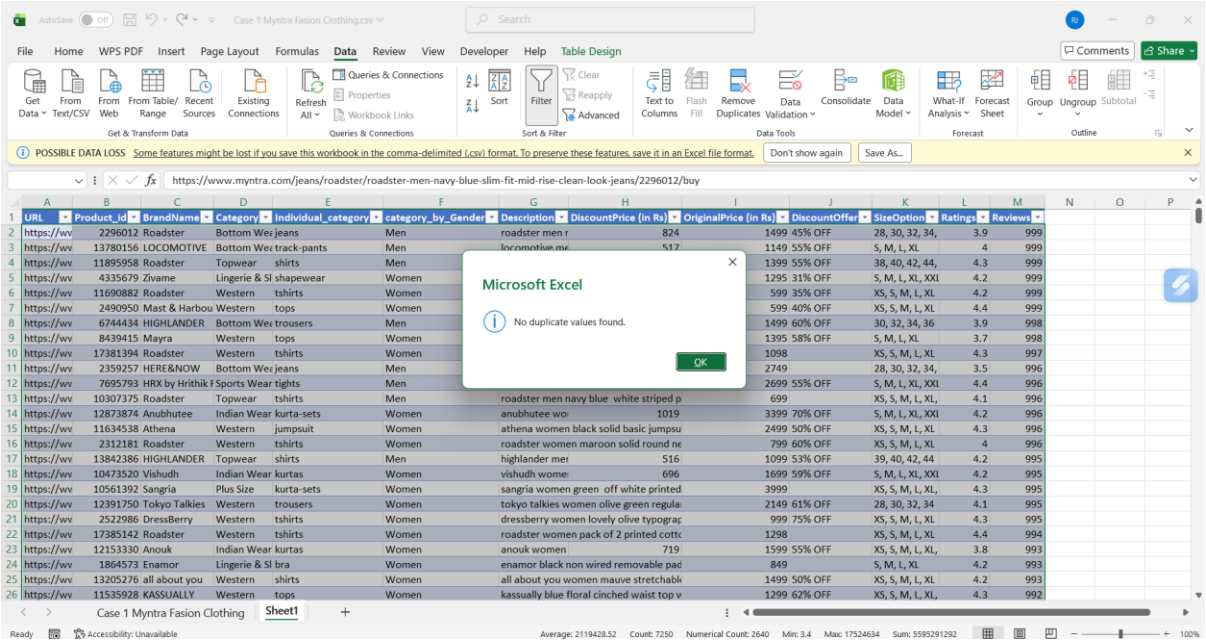
Problem Statement: You are working at Myntra, a leading online fashion retailer. The management has asked you to analyze a dataset of various apparel items to gain insights into pricing, discounts, ratings, and available sizes.

Dataset
Link: https://drive.google.com/file/d/1CDaWFvkccjdUw1E_gipTKOfMqiHNhNQL/view

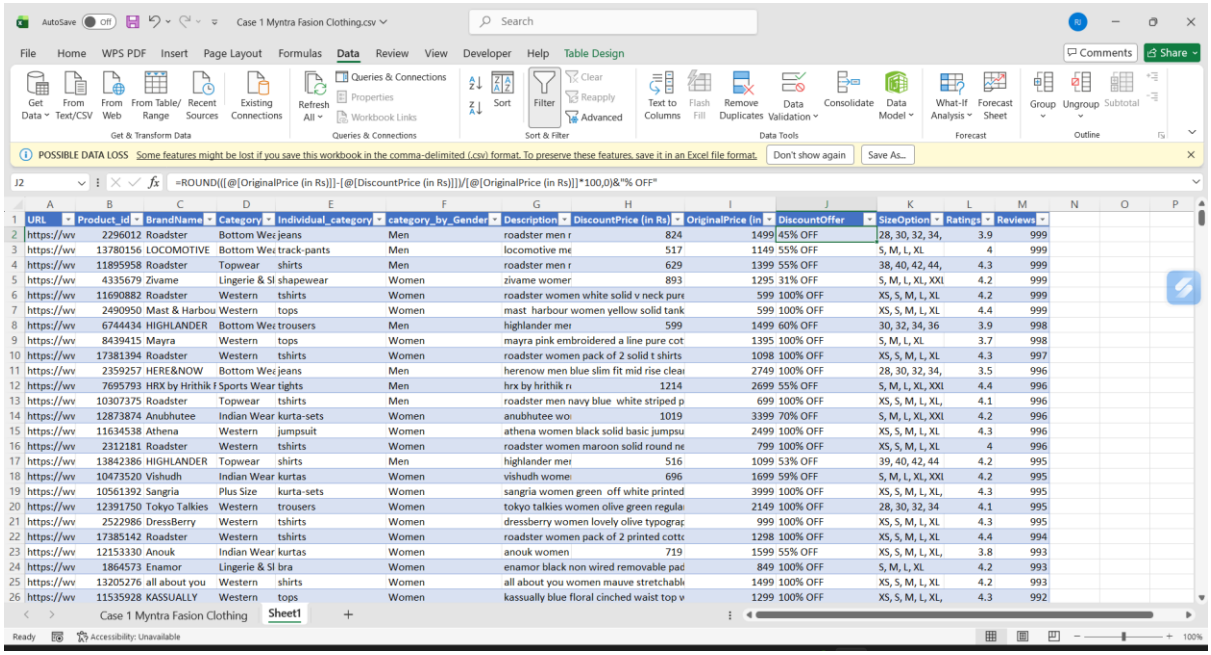
Project Questions

A. Data Cleaning and Preparation

1. Check for duplicate values in your dataset and remove them.

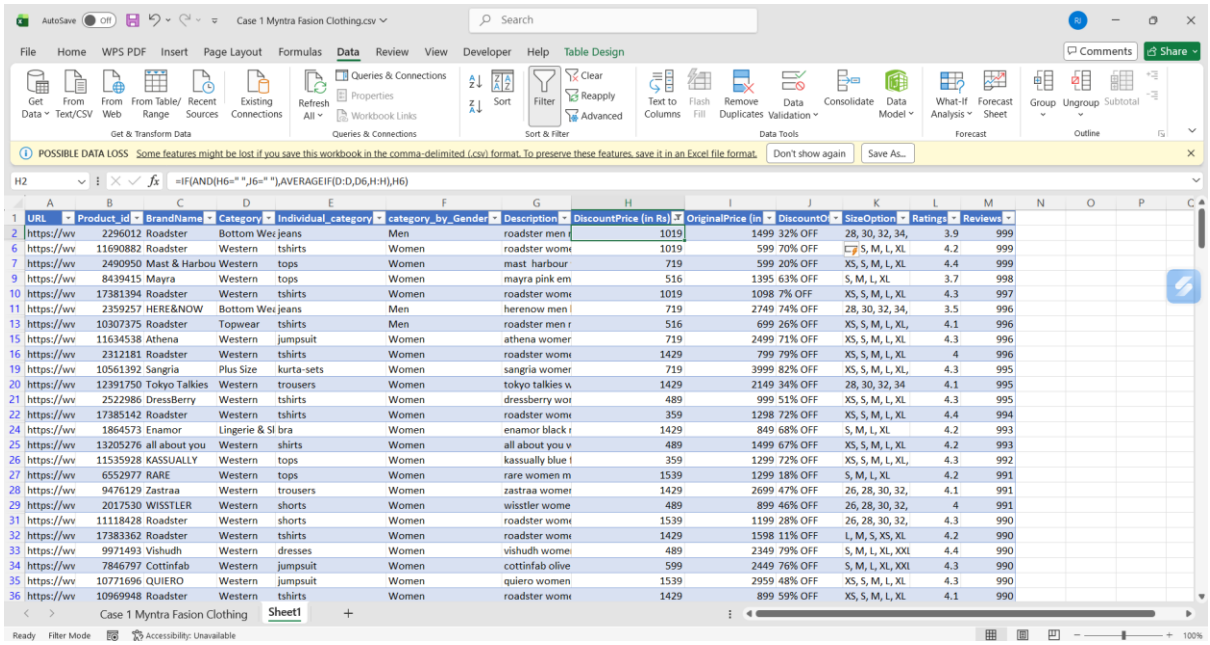


2. Standardize the "DiscountOffer" column to a single format, ensuring all values are uniform.



URL	Product_id	BrandName	Category	Individual_category	category_by_Gender	Description	DiscountPrice (in Rs)	OriginalPrice (in	DiscountOffer	SizeOption	Ratings	Reviews
https://wv	2296012	Roadster	Bottom Wea	jeans	Men	roadster men r	824	1499	45% OFF	28, 30, 32, 34,	3.9	999
https://wv	13780156	LOCOMOTIVE	Bottom Wea	track-pants	Men	locomotive me	517	1149	55% OFF	S, M, L, XL	4	999
https://wv	11895958	Roadster	Topwear	shirts	Men	roadster men r	629	1399	55% OFF	38, 40, 42, 44,	4.3	999
https://wv	4335679	Zivame	Lingerie & Sl	shapewear	Women	zivame women	893	1295	31% OFF	S, M, L, XL, XXL	4.2	999
https://wv	11690882	Roadster	Western	tshirts	Women	roadster women white solid v neck pure		599	100% OFF	XS, S, M, L, XL	4.2	999
https://wv	2490950	Mast & Harbou	Western	tops	Women	mast harbour women yellow solid tank		599	100% OFF	XS, S, M, L, XL	4.4	999
https://wv	6744434	HIGHLANDER	Bottom Wea	trousers	Men	highlander mer	599	1499	60% OFF	30, 32, 34, 36	3.9	998
https://wv	8439415	Mayra	Western	tops	Women	mayra pink embroidered a line pure cot		1395	100% OFF	S, M, L, XL	3.7	998
https://wv	17381394	Roadster	Western	tshirts	Women	roadster women pack of 2 solid t shirts		1098	100% OFF	XS, S, M, L, XL	4.3	997
https://wv	2359257	HERE&NOW	Bottom Wea	jeans	Men	herenow men blue slim fit mid rise clear		2749	100% OFF	28, 30, 32, 34,	3.5	996
https://wv	7695793	HRX by Hrithik	Sports Wear	tights	Men	hrx by hrithik r	1214	2699	55% OFF	S, M, L, XL, XXL	4.4	996
https://wv	10307375	Roadster	Topwear	tshirts	Men	roadster men navy blue white striped p		699	100% OFF	XS, S, M, L, XL,	4.1	996
https://wv	12873874	Anubhutee	Indian Wear	kurta-sets	Women	anubhutee wo	1019	3399	70% OFF	S, M, L, XL, XXL	4.2	996
https://wv	11634538	Athena	Western	jumpsuit	Women	athena women black solid basic jumpsu		2499	100% OFF	XS, S, M, L, XL	4.3	996
https://wv	2312181	Roadster	Western	tshirts	Women	roadster women maroon solid round ne		799	100% OFF	XS, S, M, L, XL	4	996
https://wv	13842386	HIGHLANDER	Topwear	shirts	Men	highlander mer	516	1099	53% OFF	39, 40, 42, 44	4.2	995
https://wv	10473520	Vishudh	Indian Wear	kurtas	Women	vishudh wome	696	1699	59% OFF	S, M, L, XL, XXL	4.2	995
https://wv	10561392	Sangria	Plus Size	kurta-sets	Women	sangria women green off white printed		3999	100% OFF	XS, S, M, L, XL,	4.3	995
https://wv	12391750	Tokyo Talkies	Western	trousers	Women	tokyo talkies women olive green regula		2149	100% OFF	28, 30, 32, 34	4.1	995
https://wv	2522986	DressBerry	Western	tshirts	Women	dressberry women lovely olive typograp		999	100% OFF	XS, S, M, L, XL	4.3	995
https://wv	17385142	Roadster	Western	tshirts	Women	roadster women pack of 2 printed cotto		1298	100% OFF	XS, S, M, L, XL	4.4	994
https://wv	12153330	Anouk	Indian Wear	kurtas	Women	anouk women	719	1599	55% OFF	XS, S, M, L, XL,	3.8	993
https://wv	1864573	Enamor	Lingerie & Sl	bra	Women	enamor black non wired removable pad		849	100% OFF	S, M, L, XL	4.2	993
https://wv	13205276	all about you	Western	shirts	Women	all about you women mauve stretchabl		1499	100% OFF	XS, S, M, L, XL	4.2	993
https://wv	11535928	KASSUALLY	Western	tops	Women	kassually blue floral cinched waist top v		1299	100% OFF	XS, S, M, L, XL,	4.3	992

3. Identify rows where both "DiscountPrice" and "DiscountOffer" are null and fill the "DiscountPrice" with the average discount price of the respective category.



URL	Product_id	BrandName	Category	Individual_category	category_by_Gender	Description	DiscountPrice (in Rs)	OriginalPrice (in	DiscountO	SizeOption	Ratings	Reviews
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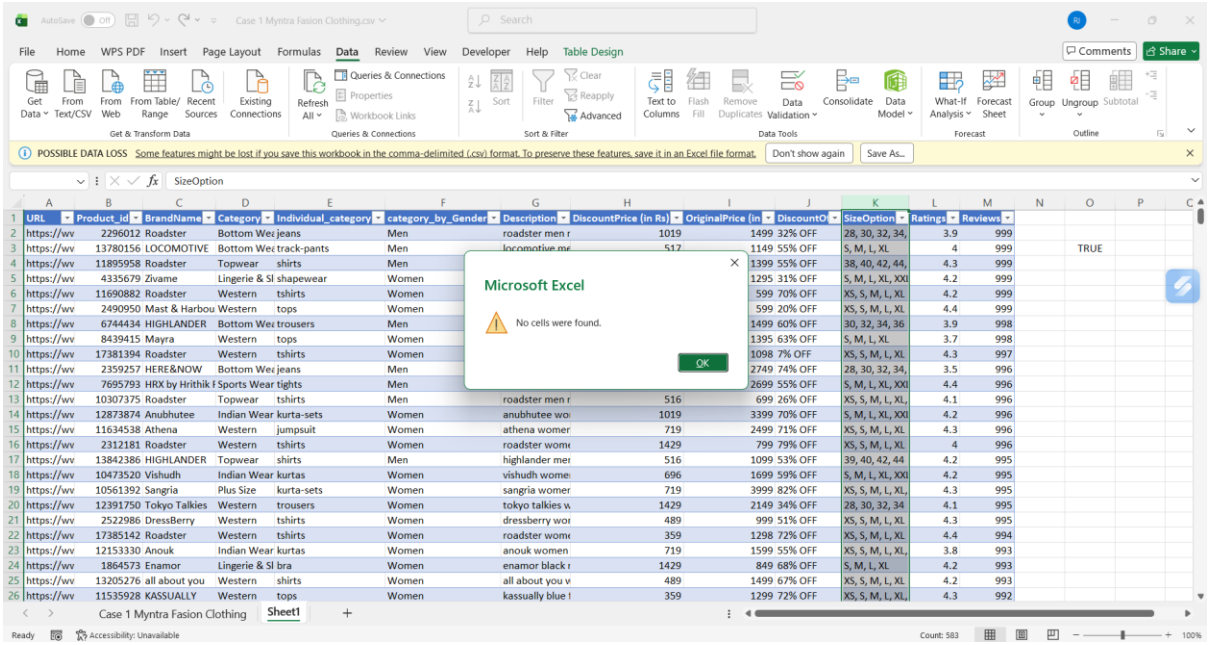
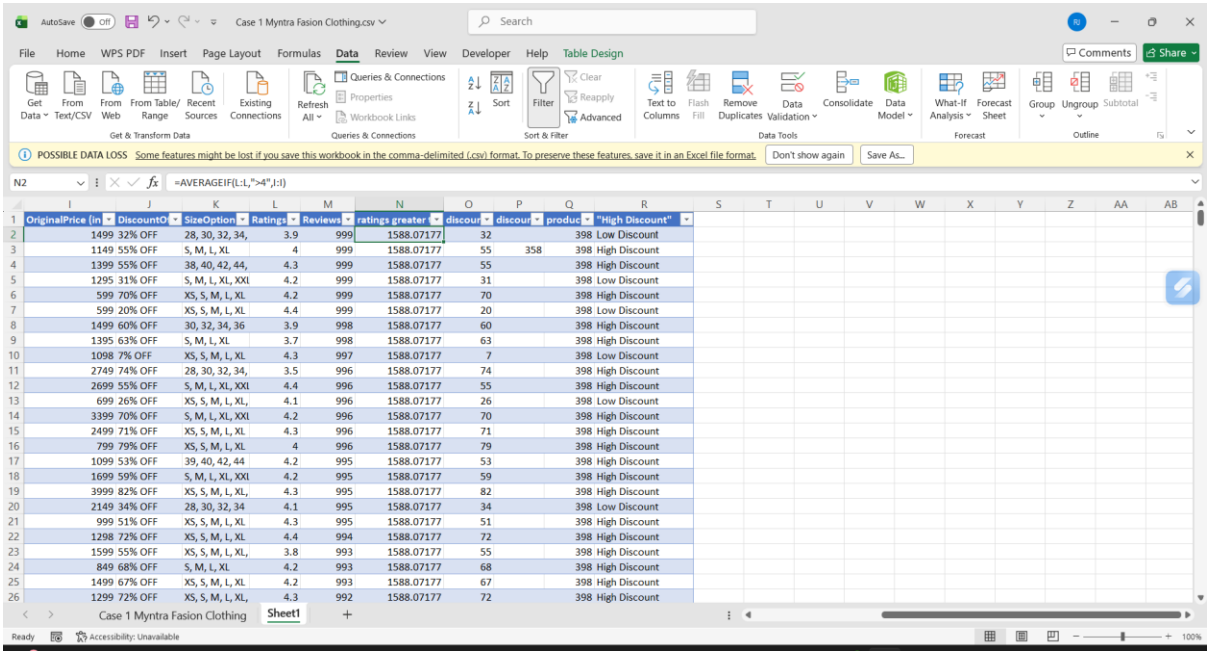
https://wv	11690882	Roadster	Western	tshirts	Women	roadster wome	1019	599	70% OFF	S, M, L, XL	4.2	999
https://wv	2490950	Mast & Harbou	Western	tops	Women	mast harbour	719	599	20% OFF	XS, S, M, L, XL	4.4	999
https://wv	8439415	Mayra	Western	tops	Women	mayra pink em	516	1395	63% OFF	S, M, L, XL	3.7	998
https://wv	17381394	Roadster	Western	tshirts	Women	roadster wome	1019	1098	7% OFF	XS, S, M, L, XL	4.3	997
https://wv	2359257	HERE&NOW	Bottom Wea	jeans	Men	herenow men	719	2749	74% OFF	28, 30, 32, 34,	3.5	996
https://wv	10307375	Roadster	Topwear	tshirts	Men	roadster men r	516	699	26% OFF	XS, S, M, L, XL,	4.1	996
https://wv	11634538	Athena	Western	jumpsuit	Women	athena women	719	2499	71% OFF	XS, S, M, L, XL	4.3	996
https://wv	2312181	Roadster	Western	tshirts	Women	roadster wome	1429	799	79% OFF	XS, S, M, L, XL	4	996
https://wv	10561392	Sangria	Plus Size	kurta-sets	Women	sangria women	719	3999	82% OFF	XS, S, M, L, XL,	4.3	995
https://wv	12391750	Tokyo Talkies	Western	trousers	Women	tokyo talkies w	1429	2149	34% OFF	28, 30, 32, 34	4.1	995
https://wv	2522986	DressBerry	Western	tshirts	Women	dressberry wor	489	999	51% OFF	XS, S, M, L, XL	4.3	995
https://wv	17385142	Roadster	Western	tshirts	Women	roadster wome	359	1298	72% OFF	XS, S, M, L, XL	4.4	994
https://wv	1864573	Enamor	Lingerie & Sl	bra	Women	enamor black	1429	849	68% OFF	S, M, L, XL	4.2	993
https://wv	13205276	all about you	Western	shirts	Women	all about you v	489	1499	67% OFF	XS, S, M, L, XL	4.2	993
https://wv	11535928	KASSUALLY	Western	tops	Women	kassually blue	359	1299	72% OFF	XS, S, M, L, XL,	4.3	992
https://wv	6552977	RARE	Western	tops	Women	rare women m	1539	1299	18% OFF	S, M, L, XL	4.2	991
https://wv	9476129	Zastraa	Western	trousers	Women	zastraa wome	1429	2699	47% OFF	26, 28, 30, 32,	4.1	991
https://wv	2017530	WISSTLER	Western	shorts	Women	wisstler wome	489	899	46% OFF	26, 28, 30, 32,	4	991
https://wv	11118428	Roadster	Western	shorts	Women	roadster wome	1539	1199	28% OFF	26, 28, 30, 32,	4.3	990
https://wv	17383362	Roadster	Western	tshirts	Women	roadster wome	1429	1598	11% OFF	L, M, S, XS, XL	4.2	990
https://wv	9971493	Vishudh	Western	dresses	Women	vishudh wome	489	2349	79% OFF	S, M, L, XL, XXL	4.4	990
https://wv	7846797	Cottinfab	Western	jumpsuit	Women	cottinfab olive	599	2449	76% OFF	S, M, L, XL, XXL	4.3	990
https://wv	10771696	QUIERO	Western	jumpsuit	Women	quiero women	1539	2959	48% OFF	XS, S, M, L, XL	4.3	990
https://wv	10969948	Roadster	Western	tshirts	Women	roadster wome	1429	899	59% OFF	XS, S, M, L, XL	4.1	990

4. Replace all null values in the "SizeOption" column with the text "Not Available."

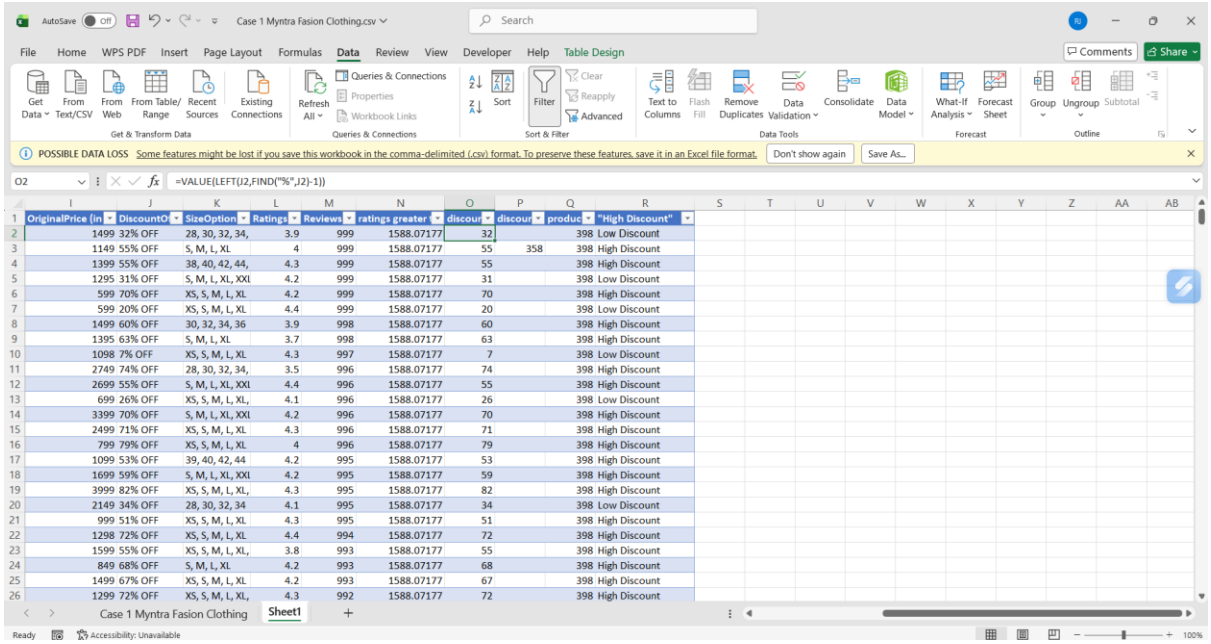


B. Data Analysis

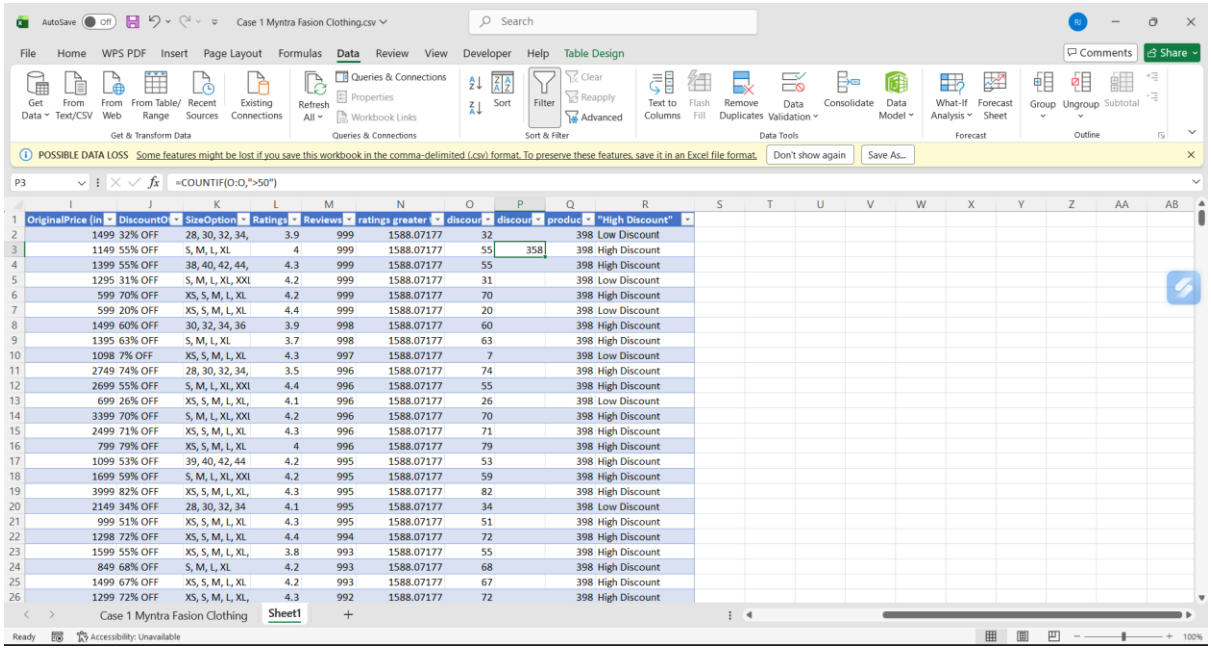
1. Calculate the overall average original price for products with ratings greater than 4.



2. Count the number of products with a discount offer greater than 50% OFF.

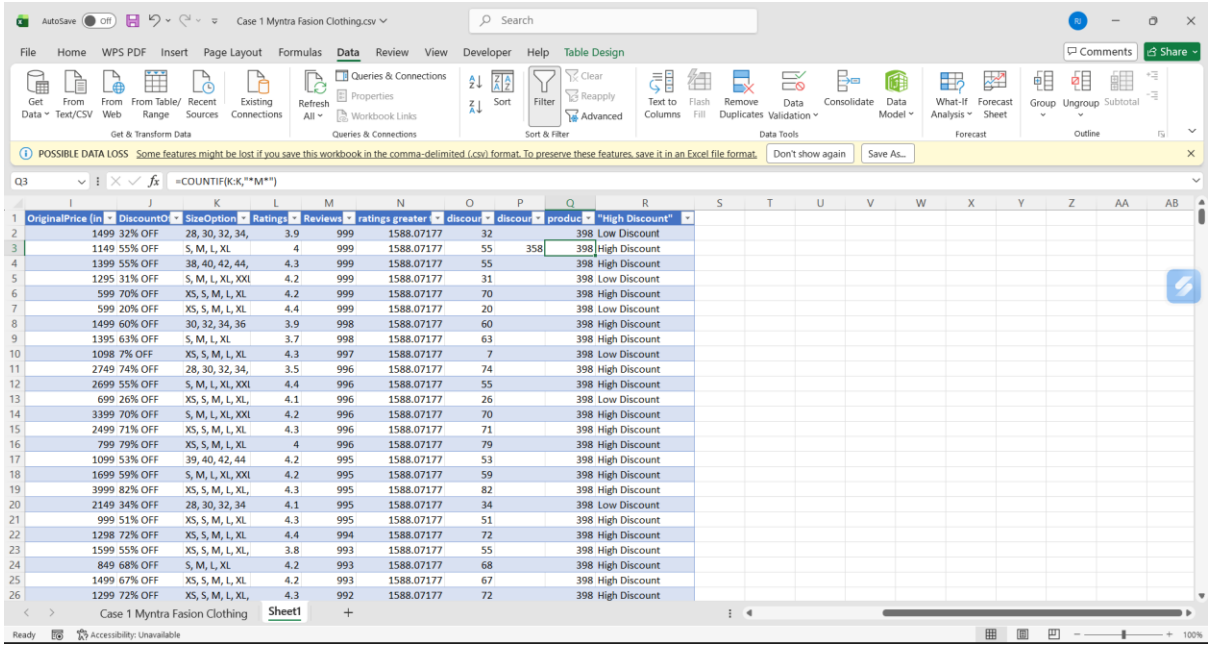


OriginalPrice (in	DiscountO	SizeOption	Ratings	Reviews	ratings greater	discour	discour	produc	"High Discount"
1499	32% OFF	28, 30, 32, 34,	3.9	999	1588.07177	32		398	Low Discount
1149	55% OFF	S, M, L, XL	4	999	1588.07177	55	358	398	High Discount
1399	55% OFF	38, 40, 42, 44,	4.3	999	1588.07177	55		398	High Discount
1295	31% OFF	S, M, L, XL, XXl	4.2	999	1588.07177	31		398	Low Discount
599	70% OFF	XS, S, M, L, XL	4.2	999	1588.07177	70		398	High Discount
599	20% OFF	XS, S, M, L, XL	4.4	999	1588.07177	20		398	Low Discount
1499	60% OFF	30, 32, 34, 36	3.9	998	1588.07177	60		398	High Discount
1395	63% OFF	S, M, L, XL	3.7	998	1588.07177	63		398	High Discount
1098	7% OFF	XS, S, M, L, XL	4.3	997	1588.07177	7		398	Low Discount
2749	74% OFF	28, 30, 32, 34,	3.5	996	1588.07177	74		398	High Discount
2699	55% OFF	S, M, L, XL, XXl	4.4	996	1588.07177	55		398	High Discount
699	26% OFF	XS, S, M, L, XL,	4.1	996	1588.07177	26		398	Low Discount
3399	70% OFF	S, M, L, XL, XXl	4.2	996	1588.07177	70		398	High Discount
2499	71% OFF	XS, S, M, L, XL	4.3	996	1588.07177	71		398	High Discount
799	79% OFF	XS, S, M, L, XL	4	996	1588.07177	79		398	High Discount
1099	53% OFF	39, 40, 42, 44	4.2	995	1588.07177	53		398	High Discount
1699	59% OFF	S, M, L, XL, XXl	4.2	995	1588.07177	59		398	High Discount
3999	82% OFF	XS, S, M, L, XL,	4.3	995	1588.07177	82		398	High Discount
2149	34% OFF	28, 30, 32, 34	4.1	995	1588.07177	34		398	Low Discount
999	51% OFF	XS, S, M, L, XL	4.3	995	1588.07177	51		398	High Discount
1298	72% OFF	XS, S, M, L, XL	4.4	994	1588.07177	72		398	High Discount
1599	55% OFF	XS, S, M, L, XL,	3.8	993	1588.07177	55		398	High Discount
849	68% OFF	S, M, L, XL	4.2	993	1588.07177	68		398	High Discount
1499	67% OFF	XS, S, M, L, XL	4.2	993	1588.07177	67		398	High Discount
1299	72% OFF	XS, S, M, L, XL,	4.3	992	1588.07177	72		398	High Discount



OriginalPrice (in	DiscountO	SizeOption	Ratings	Reviews	ratings greater	discour	discour	produc	"High Discount"
1499	32% OFF	28, 30, 32, 34,	3.9	999	1588.07177	32		398	Low Discount
1149	55% OFF	S, M, L, XL	4	999	1588.07177	55	358	398	High Discount
1399	55% OFF	38, 40, 42, 44,	4.3	999	1588.07177	55		398	High Discount
1295	31% OFF	S, M, L, XL, XXl	4.2	999	1588.07177	31		398	Low Discount
599	70% OFF	XS, S, M, L, XL	4.2	999	1588.07177	70		398	High Discount
599	20% OFF	XS, S, M, L, XL	4.4	999	1588.07177	20		398	Low Discount
1499	60% OFF	30, 32, 34, 36	3.9	998	1588.07177	60		398	High Discount
1395	63% OFF	S, M, L, XL	3.7	998	1588.07177	63		398	High Discount
1098	7% OFF	XS, S, M, L, XL	4.3	997	1588.07177	7		398	Low Discount
2749	74% OFF	28, 30, 32, 34,	3.5	996	1588.07177	74		398	High Discount
2699	55% OFF	S, M, L, XL, XXl	4.4	996	1588.07177	55		398	High Discount
699	26% OFF	XS, S, M, L, XL,	4.1	996	1588.07177	26		398	Low Discount
3399	70% OFF	S, M, L, XL, XXl	4.2	996	1588.07177	70		398	High Discount
2499	71% OFF	XS, S, M, L, XL	4.3	996	1588.07177	71		398	High Discount
799	79% OFF	XS, S, M, L, XL	4	996	1588.07177	79		398	High Discount
1099	53% OFF	39, 40, 42, 44	4.2	995	1588.07177	53		398	High Discount
1699	59% OFF	S, M, L, XL, XXl	4.2	995	1588.07177	59		398	High Discount
3999	82% OFF	XS, S, M, L, XL,	4.3	995	1588.07177	82		398	High Discount
2149	34% OFF	28, 30, 32, 34	4.1	995	1588.07177	34		398	Low Discount
999	51% OFF	XS, S, M, L, XL	4.3	995	1588.07177	51		398	High Discount
1298	72% OFF	XS, S, M, L, XL	4.4	994	1588.07177	72		398	High Discount
1599	55% OFF	XS, S, M, L, XL,	3.8	993	1588.07177	55		398	High Discount
849	68% OFF	S, M, L, XL	4.2	993	1588.07177	68		398	High Discount
1499	67% OFF	XS, S, M, L, XL	4.2	993	1588.07177	67		398	High Discount
1299	72% OFF	XS, S, M, L, XL,	4.3	992	1588.07177	72		398	High Discount

3. Count the number of products available in size "M."



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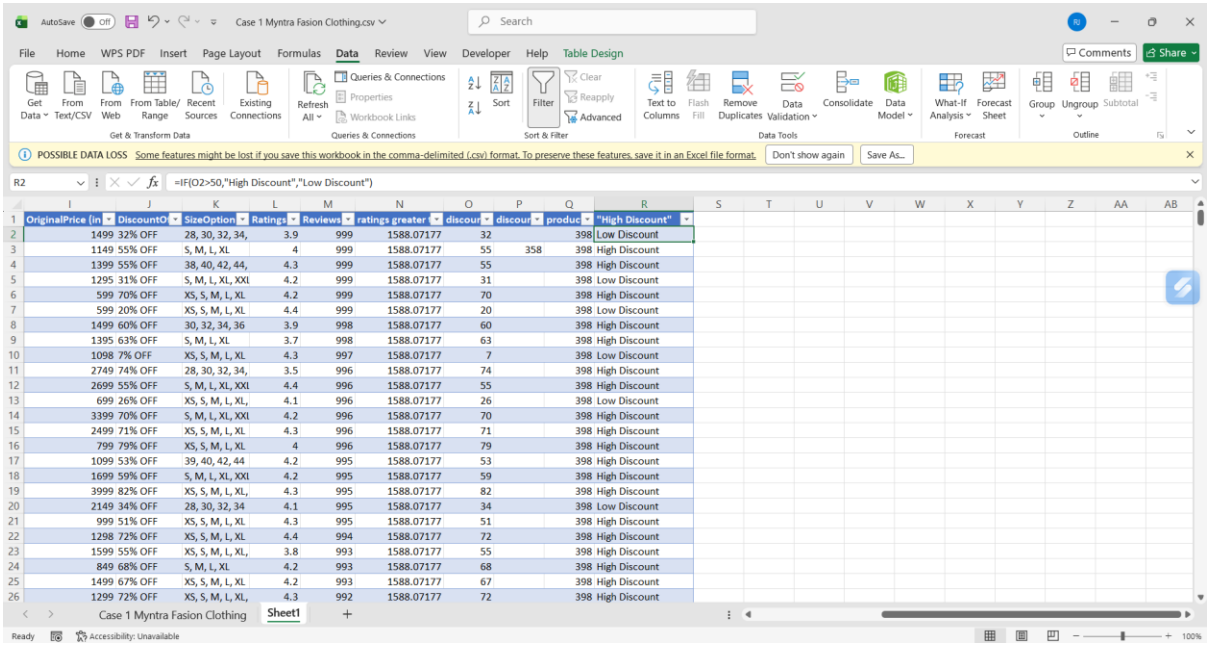
1399	55% OFF	38, 40, 42, 44,	4.3	999	1588.07177	55		398	High Discount
1295	31% OFF	S, M, L, XL, XXl	4.2	999	1588.07177	31		398	Low Discount
599	70% OFF	XS, S, M, L, XL	4.2	999	1588.07177	70		398	High Discount
599	20% OFF	XS, S, M, L, XL	4.4	999	1588.07177	20		398	Low Discount
1499	60% OFF	30, 32, 34, 36	3.9	998	1588.07177	60		398	High Discount
1395	63% OFF	S, M, L, XL	3.7	998	1588.07177	63		398	High Discount
1098	7% OFF	XS, S, M, L, XL	4.3	997	1588.07177	7		398	Low Discount
2749	74% OFF	28, 30, 32, 34,	3.5	996	1588.07177	74		398	High Discount
2699	55% OFF	S, M, L, XL, XXl	4.4	996	1588.07177	55		398	High Discount
699	26% OFF	XS, S, M, L, XL,	4.1	996	1588.07177	26		398	Low Discount
3399	70% OFF	S, M, L, XL, XXl	4.2	996	1588.07177	70		398	High Discount
2499	71% OFF	XS, S, M, L, XL	4.3	996	1588.07177	71		398	High Discount
799	79% OFF	XS, S, M, L, XL	4	996	1588.07177	79		398	High Discount
1099	53% OFF	39, 40, 42, 44	4.2	995	1588.07177	53		398	High Discount
1699	59% OFF	S, M, L, XL, XXl	4.2	995	1588.07177	59		398	High Discount
3999	82% OFF	XS, S, M, L, XL,	4.3	995	1588.07177	82		398	High Discount
2149	34% OFF	28, 30, 32, 34	4.1	995	1588.07177	34		398	Low Discount
999	51% OFF	XS, S, M, L, XL	4.3	995	1588.07177	51		398	High Discount
1298	72% OFF	XS, S, M, L, XL	4.4	994	1588.07177	72		398	High Discount
1599	55% OFF	XS, S, M, L, XL,	3.8	993	1588.07177	55		398	High Discount
849	68% OFF	S, M, L, XL	4.2	993	1588.07177	68		398	High Discount
1499	67% OFF	XS, S, M, L, XL	4.2	993	1588.07177	67		398	High Discount
1299	72% OFF	XS, S, M, L, XL,	4.3	992	1588.07177	72		398	High Discount

4. Create a new column to label the products as "High Discount" if the discount offer is greater than 50% OFF, otherwise label them as "Low Discount."



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599	20% OFF	XS, S, M, L, XL	4.4	999	1588.07177	20		398	Low Discount
1499	60% OFF	30, 32, 34, 36	3.9	998	1588.07177	60		398	High Discount
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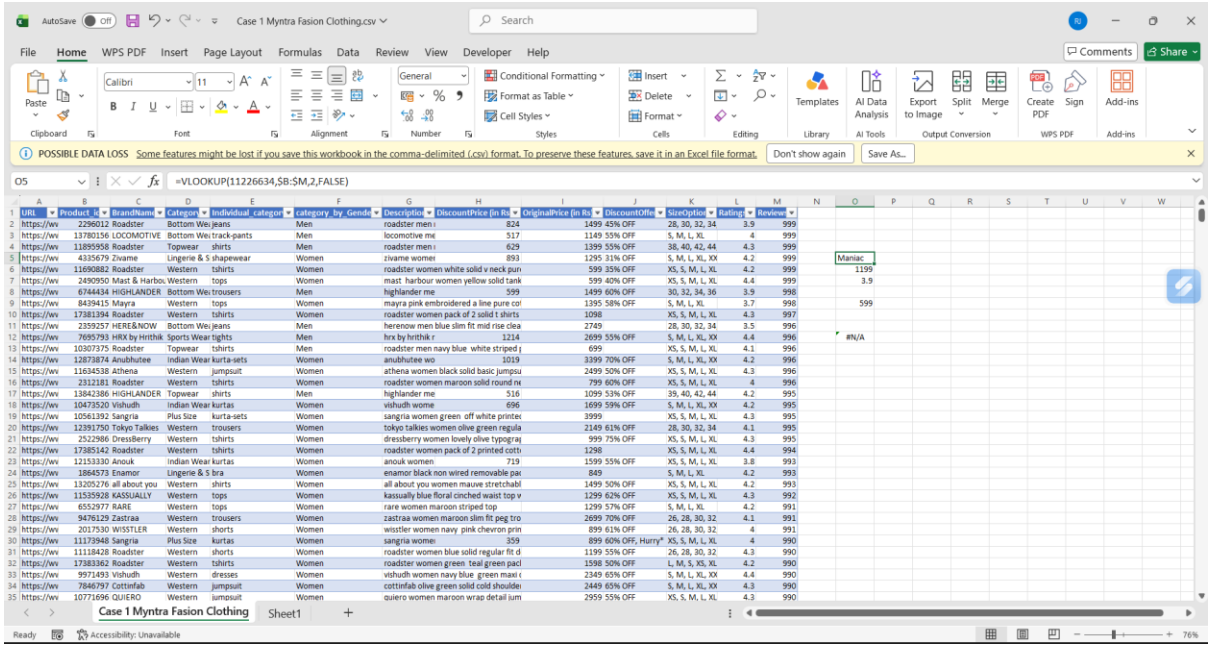
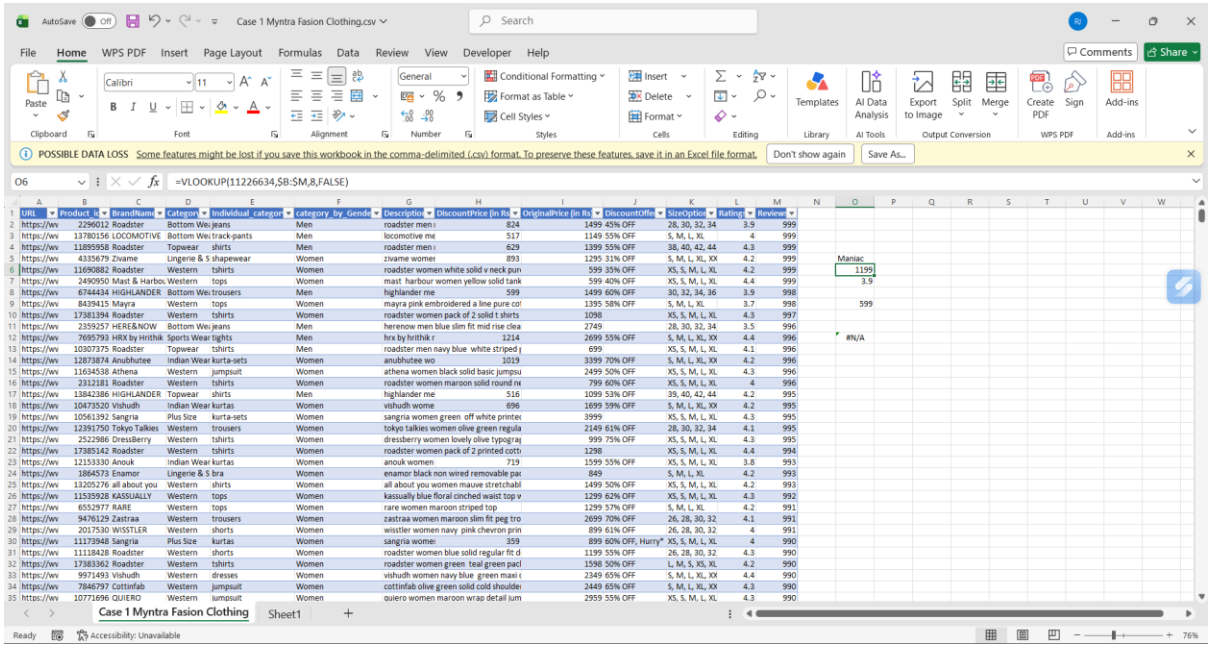
2749	74% OFF	28, 30, 32, 34,	3.5	996	1588.07177	74		398	High Discount
2699	55% OFF	S, M, L, XL, XXl	4.4	996	1588.07177	55		398	High Discount
699	26% OFF	XS, S, M, L, XL,	4.1	996	1588.07177	26		398	Low Discount
3399	70% OFF	S, M, L, XL, XXl	4.2	996	1588.07177	70		398	High Discount
2499	71% OFF	XS, S, M, L, XL	4.3	996	1588.07177	71		398	High Discount
799	79% OFF	XS, S, M, L, XL	4	996	1588.07177	79		398	High Discount
1099	53% OFF	39, 40, 42, 44	4.2	995	1588.07177	53		398	High Discount
1699	59% OFF	S, M, L, XL, XXl	4.2	995	1588.07177	59		398	High Discount
3999	82% OFF	XS, S, M, L, XL,	4.3	995	1588.07177	82		398	High Discount
2149	34% OFF	28, 30, 32, 34	4.1	995	1588.07177	34		398	Low Discount
999	51% OFF	XS, S, M, L, XL	4.3	995	1588.07177	51		398	High Discount
1298	72% OFF	XS, S, M, L, XL	4.4	994	1588.07177	72		398	High Discount
1599	55% OFF	XS, S, M, L, XL,	3.8	993	1588.07177	55		398	High Discount
849	68% OFF	S, M, L, XL	4.2	993	1588.07177	68		398	High Discount
1499	67% OFF	XS, S, M, L, XL	4.2	993	1588.07177	67		398	High Discount
1299	72% OFF	XS, S, M, L, XL,	4.3	992	1588.07177	72		398	High Discount

C. Data Retrieval and Lookup

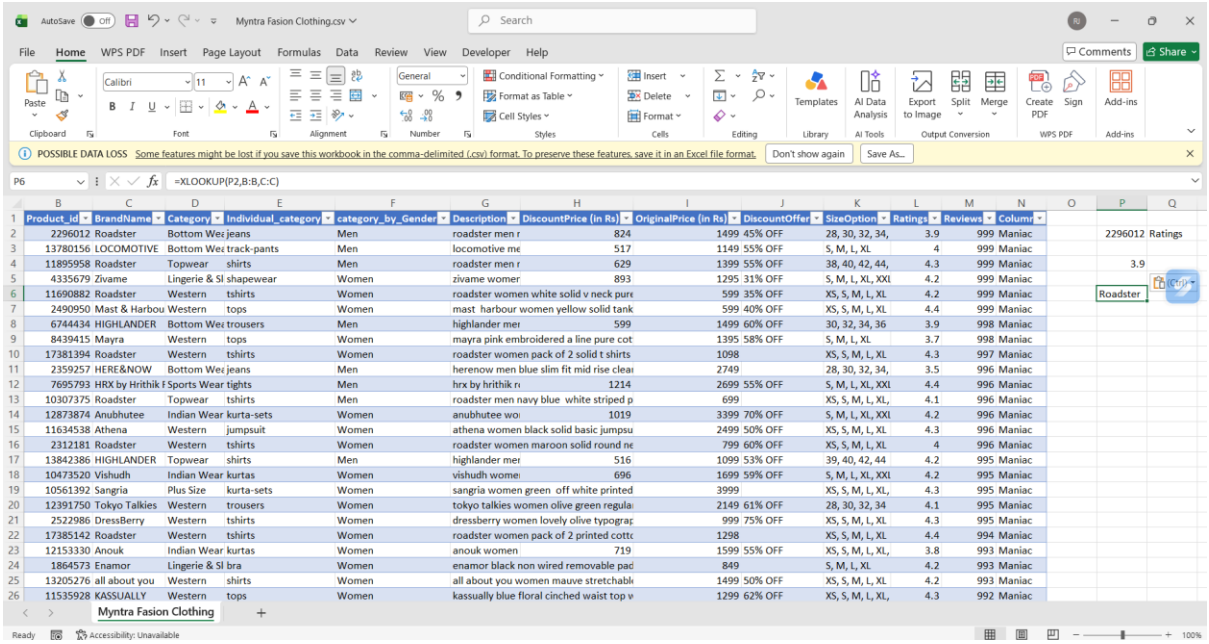
1. Use VLOOKUP/XLOOKUP to find the product brand, price, and rating of the product with Product_id "11226634".

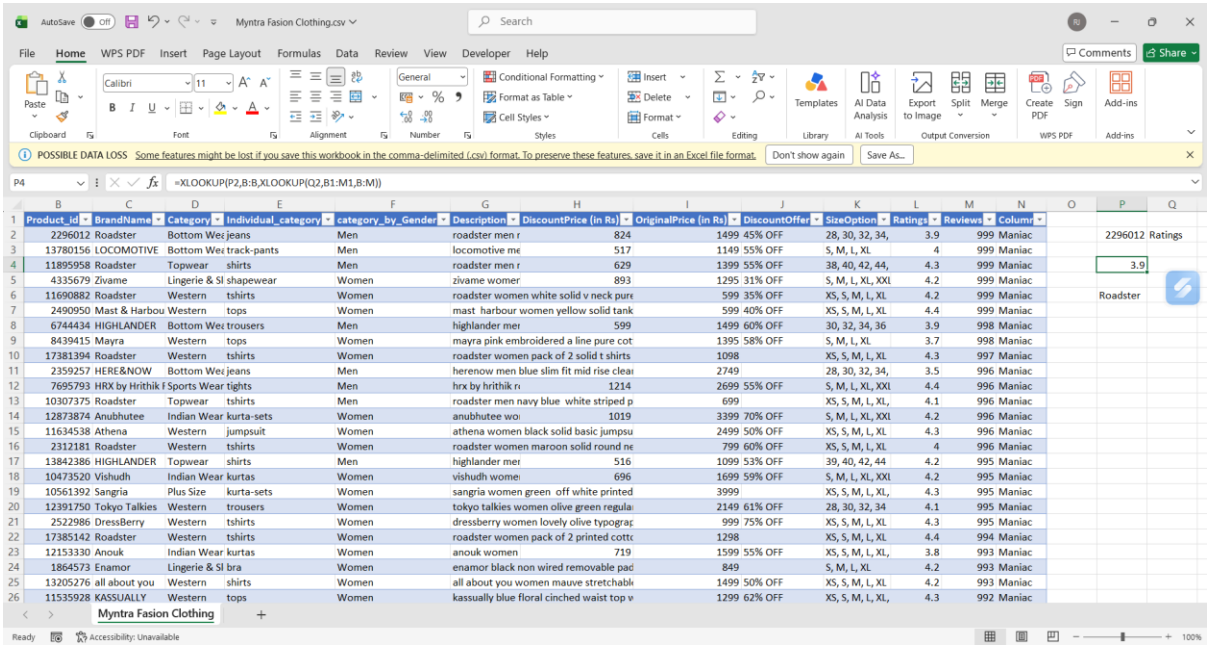


2. Find the "DiscountPrice" for the product with the Product ID "6744434" using the INDEX and MATCH functions.



3. Utilize nested xlookup to find any column’s detail of a product with it’s product id.


Myntra Fasion Clothing.csv
P6
=XLOOKUP(P2,B:B,C:C)
2296012 Ratings
3.9
Roadster
Myntra Fasion Clothing


Myntra Fasion Clothing.csv
P4
=XLOOKUP(P2,B:B,XLOOKUP(Q2,B1:M1,B:M))
2296012 Ratings
3.9
Roadster
Myntra Fasion Clothing